Problem 8.28

8.28 Water flows in a smooth 0.2 m diameter pipeline that is 65 m long. The Reynolds number is 10^6. Determine the flow rate and pressure drop. After many years of use, minerals deposited on the pipe cause the same pressure drop to produce only one-half the original flow rate. Estimate the size of the relative roughness elements for the pipe

Find: The size of the relative roughness.

Assumptions: Flow is fully developed, steady, incompressible, the kinetic energy coefficient is unity, and the fluid is Newtonian with a constant viscosity.

Solution: Use the energy equation with head loss

$$\left(\frac{p_1}{\rho g}+\frac{\bar{V}_1^2}{2g}+z_1\right)-\left(\frac{p_2}{\rho g}+\frac{\bar{V}_2^2}{2g}+z_2\right)=H_l=\frac{h_l}{g}$$

And the expression for head loss

$$H_\ell = f\,\frac{L}{D}\frac{V^2}{2g}$$

We will determine the friction factor for the original condition and then the friction factor for the deteriorated condition with the same head loss. The friction factor for the deteriorated condition will then yield the relative roughness.

For a smooth pipe, the friction factor is given by eq 8.37 with e = 0:

$$\frac{1}{\sqrt{f}}=-2.0\,log_{10}\left(\frac{2.51}{Re\,\sqrt{f}}\right)$$

With Re = $1\text{x}10^6$, the value of f is determined with an equation solver or by iteration to be

$$f\,=0.0116$$

The head loss is the same for the original and deteriorated pipes, but the deteriorated flow rate, and consequently velocity, is now one-half. Using the expression for head loss we have

$$H_\ell = f_{orig}\,\frac{L}{D}\frac{V_{orig}^2}{2g}=f_{det}\,\frac{L}{D}\frac{V_{det}^2}{2g}=f_{det}\,\frac{L}{D}\frac{\frac{V_{orig}^2}{4}}{2g}$$

Or

$$f_{det}=4f_{orig}=4\cdot 0.0116=0.0464$$

We can now use the expression for a rough pipe (eq 8.37)

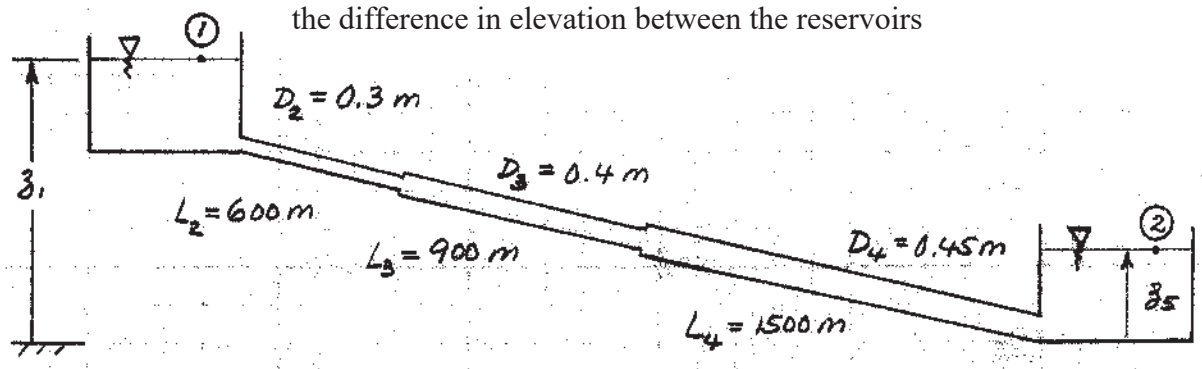
$$\frac{1}{\sqrt{f}} = -2.0\ log_{10}\left(\frac{\frac{e}{D}}{3.7} + \frac{2.51}{Re\ \sqrt{f}}\right)$$

Solving this iteratively we find

$$\frac{e}{D} = 0.0176$$

Problem 8.29

Two reservoirs are connected by three clean cast-iron pipes in series, L_1 =600 m, D_1 =0:3m, L_2 =900 m, D_2 =0:4m, L_3 =1500 m, and D_3 =0:45 m. The flow rate is 0:11 m^3/s of water at 15°C. Determine the difference in elevation between the reservoirs



<u>Solution</u>: Apply the energy equation for steady, incompressible flow that is uniform at each section.

Basic equations: $\frac{p_1}{\rho} + \frac{\bar{V}_1^2}{2} + gz_1 = \frac{p_5}{\rho} + \frac{\bar{V}_5^2}{2} + gz_5 + h_{eT}$ (1) (2) (1) (2)

$$h_{eT} = \sum f \frac{L}{D}\frac{\bar{V}^2}{2} + h_{em} \; ; \; h_{em} = K_{ent}\frac{\bar{V}_2^2}{2} + \sum h_{e_{exp}} + K_{exit}\frac{\bar{V}_4^2}{2}$$ ≈ 0(3)

Assumptions: (1) $p_1 = p_5 = p_{atm}$
(2) $\bar{V}_1 = \bar{V}_5 \simeq 0$
(3) Neglect h_{eexp} at pipe joints (note all minor losses are probably small due to long lengths of straight pipe sections, but we will check)

For non-smooth pipe, $f = f(Re, e/D)$. $\mu = 1.1 \times 10^{-3}$ N·s/m², from Table A.8.

Section ②: $e/D_2 = 0.26$ mm/300 mm = 0.00087 (for cast iron, $e = 0.26$ mm, Table 8.1)

$$\bar{V}_2 = \frac{Q}{A_2} = 0.11\frac{m^3}{s} \times \frac{4}{\pi}\frac{1}{(0.3)^2 m^2} = 1.56 \text{ m/s}$$

$$Re_2 = \frac{\rho \bar{V}_2 D_2}{\mu} = 999\frac{kg}{m^3} \times 1.56\frac{m}{s} \times 0.3\,m \times \frac{m^2}{1.14\times10^{-3} N\cdot s} \times \frac{N\cdot s^2}{kg\cdot m} = 4.10\times10^5$$

From Fig. 8.13, $f_2 = 0.020$

Section ③: $e/D_3 = 0.00065$

$$\bar{V}_3 = \frac{Q}{A_3} = 0.11\frac{m^3}{s} \times \frac{4}{\pi}\frac{1}{(0.4)^2 m^2} = 0.875 \text{ m/s}$$

$$Re_3 = \frac{\rho \bar{V}_3 D_3}{\mu} = 999\frac{kg}{m^3} \times 0.875\frac{m}{s} \times 0.4\,m \times \frac{m^2}{1.14\times10^{-3} N\cdot s} \times \frac{N\cdot s^2}{kg\cdot m} = 3.07\times10^5$$

From Fig. 8.13, $f_3 = 0.019$

Section ④: $e/D_4 = 0.00058$

$$\overline{V}_4 = \frac{Q}{A_4} = 0.11\,\frac{m^3}{s} \times \frac{4}{\pi}\,\frac{1}{(0.45)^2 m^2} = 0.692\ m/s$$

$$Re_4 = \frac{\rho \overline{V}_4 D_4}{\mu} = 999\,\frac{kg}{m^3} \times 0.692\,\frac{m}{s} \times 0.45\,m \times \frac{m^2}{1.14\times10^{-3}\,N\cdot s} \times \frac{N\cdot s^2}{kg\cdot m} = 2.73\times10^5$$

From Fig. 8.13, $f_4 = 0.0185$

Then $$\sum f \frac{L}{D}\frac{\overline{V}^2}{2} = 0.020 \times \frac{600\,m}{0.3\,m} \times \frac{1}{2}(1.56)^2\frac{m^2}{s^2} + 0.019 \times \frac{900\,m}{0.4\,m} \times \frac{1}{2}(0.875)^2\frac{m^2}{s^2}$$

$$+ 0.0185 \times \frac{1500\,m}{0.45\,m} \times \frac{1}{2}(0.692)^2\frac{m^2}{s^2} = 79.8\ m^2/s^2$$

The minor loss coefficients are $K_{ent} = 0.5$ (Table 8.2) and $K_{exit} = 1.0$. Thus,

$$h_{em} = K_{ent}\frac{\overline{V}_2^2}{2} + K_{exit}\frac{\overline{V}_4^2}{2}$$

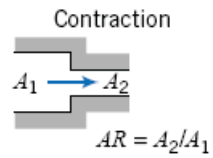
$$h_{em} = 0.5 \times \frac{1}{2} \times (1.56)^2\frac{m^2}{s^2} + 1.0 \times \frac{1}{2} \times (0.692)^2\frac{m^2}{s^2} = 0.848\ m^2/s^2$$

Therefore minor losses are roughly 1 percent of the frictional losses, so they may be neglected. Thus from the energy equation

$$z_1 - z_5 = \sum f \frac{L}{D}\frac{\overline{V}^2}{2g} = 79.8\,\frac{m^2}{s^2} \times \frac{s^2}{9.81\,m} = 8.13\ m$$ $z_1 - z_5$

Problem 8.31

8.31 Water flows through a 2-in.-diameter tube that suddenly contracts to 1 in. diameter. The pressure drop across the contraction is 0.5 psi. Determine the volume flow rate.



Given: Flow through sudden contraction

Find: Volume flow rate

Solution:

Basic equations $\left(\frac{p_1}{\rho} + \alpha\cdot\frac{V_1^2}{2} + g\cdot z_1\right) - \left(\frac{p_2}{\rho} + \alpha\cdot\frac{V_2^2}{2} + g\cdot z_2\right) = h_{lm}$ $\quad h_{lm} = K\cdot\frac{V_2^2}{2}$ $\quad Q = V\cdot A$

Assumptions: 1) Steady flow 2) Incompressible flow 3) α at 1 and 2 is approximately 1 4) Horizontal

Hence the energy equation becomes

$$\left(\frac{p_1}{\rho} + \frac{V_1^2}{2}\right) - \left(\frac{p_2}{\rho} + \frac{V_2^2}{2}\right) = K\cdot\frac{V_2^2}{2}$$

From continuity $\quad V_1 = V_2\cdot\frac{A_2}{A_1} = V_2\cdot AR$

Hence $\quad \left(\frac{p_1}{\rho} + \frac{V_2^2\cdot AR^2}{2}\right) - \left(\frac{p_2}{\rho} + \frac{V_2^2}{2}\right) = K\cdot\frac{V_2^2}{2}$

Solving for V_2 $\quad V_2 = \sqrt{\frac{2\cdot(p_1 - p_2)}{\rho\cdot(1 - AR^2 + K)}}$ $\quad AR = \left(\frac{D_2}{D_1}\right)^2 = \left(\frac{1}{2}\right)^2 = 0.25$ $\quad$ so from Fig. 8.14 $\quad K = 0.4$

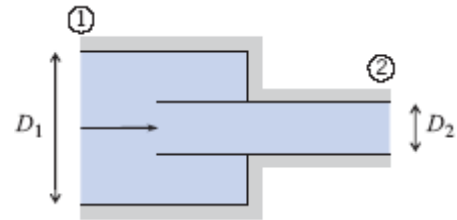
Hence $\quad V_2 = \sqrt{2 \times 0.5\cdot\frac{lbf}{in^2} \times \left(\frac{12\cdot in}{1\cdot ft}\right)^2 \times \frac{ft^3}{1.94\cdot slug} \times \frac{1}{(1 - 0.25^2 + 0.4)} \times \frac{slug\cdot ft}{lbf\cdot s^2}}$ $\quad V_2 = 7.45\cdot\frac{ft}{s}$

$$Q = V_2\cdot A_2 = \frac{\pi\cdot D_2^2}{4}\cdot V_2$$

$Q = \frac{\pi}{4} \times \left(\frac{1}{12}\cdot ft\right)^2 \times 7.45\cdot\frac{ft}{s}$ $\quad Q = 0.0406\cdot\frac{ft^3}{s}$ $\quad Q = 2.44\cdot\frac{ft^3}{min}$ $\quad Q = 18.2\cdot gpm$

Problem 8.33

8.33 Water flows from a larger pipe, diameter $D_1 = 100\,mm$, into a smaller one, diameter $D_2 = 50\,mm$, by way of a reentrant device. Find the head loss between points ① and ②.



Given: Flow through a reentrant device

Find: Head loss

Solution:

Basic equations $\left(\frac{p_1}{\rho} + \alpha\cdot\frac{V_1^2}{2} + g\cdot z_1\right) - \left(\frac{p_2}{\rho} + \alpha\cdot\frac{V_2^2}{2} + g\cdot z_2\right) = h_{lT}$ $\quad h_{lT} = h_l + h_{lm} = f\cdot\frac{L}{D}\cdot\frac{V_2^2}{2} + K\cdot\frac{V_2^2}{2}$ $\quad Q = V\cdot A$

Assumptions: 1) Steady flow 2) Incompressible flow 3) α at 1 and 2 is approximately 1 4) L << so ignore h_l 5) Reentrant

Available data $D_1 = 100\cdot mm$ $\quad D_2 = 50\cdot mm$ $\quad Q = 0.01\cdot\frac{m^3}{s}$ and from Table 8.2 $\quad K = 0.78$

$A_1 = \frac{\pi}{4}\cdot D_1^2$ $\quad A_1 = 7.85\times 10^3\,mm^2$ $\quad A_2 = \frac{\pi}{4}\cdot D_2^2$ $\quad A_2 = 1.96\times 10^3\,mm^2$

Hence between the free surface (Point 1) and the exit (2) the energy equation becomes

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} - \frac{V_2^2}{2} - \frac{p_2}{\rho} = K\cdot\frac{V_2^2}{2}$$

From continuity $Q = V_1\cdot A_1 = V_2\cdot A_2$ and also $\frac{p_1 - p_2}{\rho} = \frac{\rho\cdot g\cdot h}{\rho} = g\cdot h$ where h is the head loss

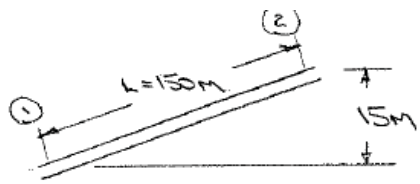
Hence $$g\cdot h + \frac{1}{2}\cdot\left(\frac{Q}{A_1}\right)^2 - \frac{1}{2}\cdot\left(\frac{Q}{A_2}\right)^2 = K\cdot\frac{1}{2}\cdot\left(\frac{Q}{A_2}\right)^2$$

Solving for h $$h = \frac{\left(\frac{Q}{A_2}\right)^2}{2\cdot g}\cdot\left[1 + K - \left(\frac{A_2}{A_1}\right)^2\right]$$ $\quad h = 2.27\,m$

Problem 8.38

8.38 Water flows steadily in a 125-mm-diameter cast-iron pipe 150 m long. The pressure drop between sections (1) and (2) is 150 kPa , and section (2) is located 15 m above section (1). Find the volume flow rate.

Assumptions: (1) $\bar{V}_1 = \bar{V}_2$, from continuity (2) $\alpha_1 = \alpha_2$ (3) $z_2 - z_1 = 15\ m$ (4) neglect minor losses



Solution:

Apply the energy equation with head loss for steady incompressible pipe flow.

$$\left(\frac{p_1}{\rho} + \alpha_1 \frac{\bar{V}_1^2}{2} + gz_1\right) - \left(\frac{p_2}{\rho} + \alpha_2 \frac{\bar{V}_2^2}{2} + gz_2\right) = h_{lT} \tag{1}$$

$$h_{lT} = h_l + h_{lm} = f\frac{L}{D}\frac{\bar{V}^2}{2} + K\frac{\bar{V}^2}{2} \tag{2}$$

For cast iron pipe with:

$$D = 125\ mm$$

$$e = 0.26\ mm$$

$$\frac{e}{D} = 0.0021$$

Since $f = f(Re)$ and $\bar{V}$ is unknown, iteration will be required:

From Eqs (1) and (2):

$$\left(\frac{p_1}{\rho} + gz_1\right) - \left(\frac{p_2}{\rho} + gz_2\right) = f\frac{L}{D}\frac{\bar{V}^2}{2}$$

Then

$$f\bar{V}^2 = \frac{2D}{L}\left[\frac{(p_1 - p_2)}{\rho} + g(z_1 - z_2)\right]$$

$$f\bar{V}^2 = \frac{2 \times 0.125\ m}{150\ m} \times \left[\frac{150 \times 10^3\ \frac{N}{m^2}}{999\ \frac{kg}{m^3}} + 9.81\ \frac{m}{s^2} \times (-15\ m)\right] = 0.005\ \frac{m^2}{s^2}$$

Assume flow in fully rough region, $f = 0.0237$, then $\bar{V} = 0.46\ \frac{m}{s}$.

Check Re assuming $T = 15\ ℃$, $\nu = 1.14 \times 10^{-6}\ \frac{m^2}{s}$,

$$Re = \frac{D\bar{V}}{\nu} = \frac{0.125\ m \times 0.46\ \frac{m}{s}}{1.14 \times 10^{-6}\ \frac{m^2}{s}} = 50400$$

From Eq. 8.37, with $Re = 50400$, $e/D = 0.0021$, then using Excel or an equation solver:

$$f = 0.0267$$

$$\bar{V} = 0.433\ \frac{m}{s}$$

With this value of $\bar{V}$, $Re = 47500$,

$$f = 0.0268$$

$$\bar{V} = 0.432\ \frac{m}{s}$$

Then the flow rate is

$$Q = A\bar{V} = \frac{\pi D^2}{4}\bar{V} = \frac{\pi}{4} \times (0.125\ m)^2 \times 0.432\ \frac{m}{s} = 0.0053\ \frac{m^3}{s}$$

Problem 8.43

8.43 An industrial plant requires a water flow rate of 5.7m^3/min. The gage pressure in the water main, located in the street 50 m from the plant, is 800 kPa and the gage pressure required in the plant is 500 kPa. The supply line will require installation of 4 elbows in a total length of 65 m. Determine the smallest size of galvanized iron line that should be installed.

Assumptions: (1) $p_1 - p_2 \leq 300\ kPa = \Delta p$ (2) fully developed flow in constant-area pipe, $\bar{V}_1 = \bar{V}_2 = \bar{V}$ (3) $z_1 = z_2$ (4) $h_{lm} = 4\left(\frac{L_e}{D}\right)_{elbow}\frac{\bar{V}^2}{2} = 120\frac{\bar{V}^2}{2} \quad \left(\frac{L_e}{D} = 30\right)$

Solution: Apply the energy equation with head loss for steady, incompressible flow that is uniform at each section ($\alpha \approx 1$).

Basic equation:

$$\left(\frac{p_1}{\rho} + \alpha_1\frac{\bar{V}_1^2}{2} + gz_1\right) = \left(\frac{p_2}{\rho} + \alpha_2\frac{\bar{V}_2^2}{2} + gz_2\right) + f\frac{L}{D}\frac{\bar{V}^2}{2} + h_{lm}$$

Then

$$\frac{\Delta p}{\rho} = f\left(\frac{L}{D} + 120\right)\frac{\bar{V}^2}{2}$$

or

$$\Delta p = \rho f\left(\frac{L}{D} + 120\right)\frac{\bar{V}^2}{2}$$

Since D is unknown, iteration is required. The calculation equations are:

$$\bar{V} = \frac{Q}{A} = \frac{4Q}{\pi D^2} = \frac{4 \times \frac{5.7}{60}\frac{m^3}{s}}{\pi \times D^2\, m^2} = \frac{0.121}{D^2}\frac{m}{s}$$

$$Re = \frac{\bar{V}D}{\upsilon} = \frac{4Q}{\pi\upsilon D} = \frac{4 \times \frac{5.7}{60}\frac{m^3}{s}}{\pi \times 1.14 \times 10^{-6}\frac{m^2}{s} \times D\, m} = \frac{1.06 \times 10^5}{D}$$

$e = 0.15\ mm$ (Table 8.1), f from Eq. 8.37, $L = 65\ m$. D from Table 8.5:

D (nom)	D (m)	$\bar{V}$ (m/s)	Re $(-)$	e/D $(-)$	f $(-)$	L/D $(-)$	Δp (kPa)
3	0.0779	19.9	1.36×10^6	0.0019	0.024	834	4530
5	0.128	7.39	8.29×10^5	0.0012	0.021	508	360
6	0.154	5.10	6.89×10^5	0.001	0.020	422	141

Pipe friction calculations are accelerated only within about ± 10 percent. Line resistance (and consequently Δp) will increase with age.

Recommend installation of 6 in (nominal) line.

Problem 8.46

8.46 Determine the diameter of a 175 m long pipe with a roughness of 2.5mm that will provide a flow of 0.075 m^3/s with a 500 kPa pressure drop.

Find: The pipe diameter

Assumptions: Flow is fully developed, steady, and incompressible and the kinetic energy coefficient is unity

Solution: Use the energy equation with head loss and the expressions for pipe friction loss

$$\left(\frac{p_1}{\rho}+\frac{\bar{V}_1^2}{2}+gz_1\right)-\left(\frac{p_2}{\rho}+\frac{\bar{V}_2^2}{2}+gz_2\right)=h_l=H_l g$$

With the head loss given by:

$$H_l=f\frac{L}{D}\frac{\bar{V}^2}{2g}$$

We can simplify the energy equation as the pipe is uniform in diameter (constant velocity) and horizontal:

$$\frac{p_1-p_2}{\rho}=H_l g$$

With the expression for head loss:

$$\frac{p_1-p_2}{\rho}=f\frac{L}{D}\frac{\bar{V}^2}{2}$$

Or

$$f\frac{L}{D}\frac{\bar{V}^2}{2}=\frac{(p_1-p_2)}{\rho}=\frac{500kPa}{1000\ \frac{kg}{m^3}}$$

The velocity and diameter are related to the volume flow rate through the continuity relation

$$Q=VA=V\cdot\frac{\pi}{4}\cdot D^2=0.075\ \frac{m^3}{s}$$

The friction factor is related to the velocity and diameter through the empirical relation eq. 8.37:

$$\frac{1}{\sqrt{f}} = -2\,log\left(\frac{e/D}{3.7} + \frac{2.51}{Re\,\sqrt{f}}\right)$$

Where the Reynolds number is given by

$$Re = \frac{V\,\rho\,D}{\mu} = V\,D\,\frac{1000\,\frac{kg}{m^3}}{0.00101\,\frac{kg}{m-s}}$$

These are four coupled equations with the four unknown variables V, D, f, and Re. Using iteration or an equation solver, the solution is obtained as:

$$D = 0.150\ m$$

$$f = 0.0456$$

$$Re = 641{,}000$$

$$V = 4.32\ m/s$$